



Tertiary Infotech Academy Pte Ltd

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LAB 01 INSTRUCTIONS

For

Establish a trace baseline

Course Code: C1123

Conducted by

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Version 5.1

Lab 01 — Establish a trace baseline

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 01 — Introduction to Network Analysis and Wireshark

Learning outcome: LO1 — Plan scoped captures and configure a reproducible analyst profile.

Time: about 40 minutes

Version: v5.1 · 5 October 2026

Scenario

The branch help desk has received a capture from the client PC (192.0.2.10), but nobody has checked what it contains. Before anyone diagnoses a fault, you establish a trusted baseline of the trace.

Goal

A capture is a measurement at one observation point, not a complete network history.

What you will produce

A baseline record of the capture and the ARP exchange.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to outputs/evidence.csv
outputs/findings.md	Findings sheet you complete

Lab at a glance

1. Verify the lab fixture with `verify.py`
2. Record packet count and capture duration
3. Trace a field from Packet Details to bytes
4. Pair the ARP request and reply

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `data/branch-office.pcap` with `File > Open`. Clear the display filter. Record the packet count from the status bar and the capture duration from `Statistics > Capture File Properties`.
3. Select frame 1. Expand Ethernet II and Address Resolution Protocol in Packet Details. Click the sender IP field and observe the corresponding bytes.
4. Apply `arp` in the display filter toolbar. Compare request and reply sender/target IP and MAC addresses. Save the two frame numbers in `outputs/findings.md`.
5. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

arp					
No.	Time	Source	Destination	Protocol	Info
1	0.000000	02:00:00:00:00:10	ff:ff:ff:ff:ff:ff	ARP	Who has 192.0.2.20? Tell 192.0.2.10
2	0.010000	02:00:00:00:00:20	02:00:00:00:00:10	ARP	192.0.2.20 is at 02:00:00:00:00:20
branch-office.pcap Displayed: 2					

Lab 01 expected evidence

Test it

The supplied `arp` expression matches 2 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in `outputs/`. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.
- TLS remains opaque: select the matching `lab-tls.keys` file by absolute path, reload, and remove a `key-log` preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -Y arp
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

Getting Started lab — capture your own short baseline on an authorised network and compare its protocol mix with this file. Source: J.F. Kurose and K.W. Ross, Wireshark Labs (gaia.cs.umass.edu/kurose_ross/wireshark.php).

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.